K.T.G.Tharusha Nimesh Dhananjaya

B.Sc. Electronics & Telecommunication Engineering (Undergraduate)

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TharushaNimesh22

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Education

University of Moratuwa, Sri Lanka 2024 – present
B.Sc. Eng. Hons. in Electronic & Telecommunication Engineering

- **CGPA – 3.99/4.00**
- **Dean's List – 1, 2, 3, 4 Semesters**
- **Coursework** – Electronics, Signals and Systems, Pattern Recognition, Computer Vision, Data Structures and Algorithms, Control Systems, Digital Design, Telecommunication, Linear Algebra

Maliyadeva College, Kurunegala 2020 – 2022
G.C.E. Advanced Level Examination 2022(2023) – Physical Science Stream

- Combined Mathematics, Physics, Chemistry – 3A's
- Z Score – (+2.382) / District Rank – 27 / Island Rank – 352

Wayamba Royal College, Kurunegala 2014 – 2019
G.C.E. Ordinary Level Examination 2019

- Buddhism, Sinhala Language & Literature, Mathematics, Science, English, History, Tamil, Sinhala Literature, Information and Communication Technology – 9A's

Courses & Certifications

1. Sri Lanka Robotics Challenge (SLRC) 2026 – Finalist

2. Machine Learning Specialization – Coursera (Offered by DeepLearning.AI + Stanford University)
   - **Coursework** – Supervised Machine Learning (Regression and Classification), Advanced Learning Algorithms (Neural Networks), Unsupervised Machine Learning, Recommender Systems and Reinforcement Learning.

3. Deep Learning Specialization – Coursera (Offered by DeepLearning.AI) In Progress
   - **Coursework** – Neural Networks, Hyperparameter Tuning, Regularization and Optimization, Convolutional Neural Networks, Sequence Models

Skills

- **Languages** – Sinhalese, English
- **Programming Languages** – Python, C, C++, MATLAB, JavaScript
- **Machine Learning Libraries** – NumPy, Scikit-learn, PyTorch, TensorFlow & Pandas
- **Hardware** – System Verilog HDL, FPGA, STM32, Arduino, PIC
- **CAD & EDA Tools** – Solidworks & Altium PCB Designer
- **Documentation** – LaTeX

Projects

Wearable ECG Monitoring System July 2026
(Electronics, IoT, Signal Processing, CNN, Embedded Systems)

- Developed a compact wearable ECG device capable of long-term ambulatory cardiac monitoring, with a focus on early detection of abnormalities in high-risk individuals.
- Built an embedded analytics pipeline that combines real-time ECG signal conditioning with CNN-based pattern recognition to support faster clinical assessment.
- Designed, implemented and evaluated three deep learning architectures (ResNet, ResNet + BiLSTM, and ResNet + Attention) to identify the optimal model for the application.

Brain Tumor Classification Using CNNs September 2026
(Deep Learning, CNNs, PyTorch, Medical Image Classification)

- Developed and evaluated CNN architectures using PyTorch to classify brain MRI images into four categories (glioma, meningioma, pituitary tumor and no tumor).
- Compared a conventional CNN with a lightweight architecture using depthwise separable convolutions and evaluated the impact of different optimization algorithms (Adam, SGD and SGD with Momentum) on classification performance.

Neural Networks for Computer Vision October 2026
(Deep Learning, CNNs, Transfer Learning, PyTorch, Computer Vision)

- Implemented and evaluated neural network architectures for image classification including a fully connected network, LeNet-5 and a modern CNN using PyTorch.
- Applied transfer learning with pre-trained ResNet-18 for ant and bee classification, comparing fine-tuning against feature extraction and achieving up to 94.77% validation accuracy with only 1,026 trainable parameters.

Atom – Robotics Design December 2025
(Robotics, Control Systems, Embedded Systems, C++)

- Designed and developed an autonomous mobile robot capable of line and dashed-line following, grid navigation, arrow following, barcode reading, color recognition, wall following, ramp traversal, and autonomous object manipulation for a multi-stage robotics competition.
- Developed the robot control software in C++ on an Arduino Mega platform, implementing PID-based line-following and wall-following controllers and integrating multiple sensors, stepper motors, servo mechanisms and custom PCB electronics.

UART Transceiver Implementation in FPGA April 2026
(FPGA, Verilog)

- Designed and implemented a UART transceiver in Verilog HDL on FPGA, including RTL development of transmitter, receiver, and top-level modules for asynchronous serial communication.
- Developed comprehensive testbench for functional verification and validated the design on FPGA hardware through successful bidirectional communication between two FPGA boards.

Three Channel Audio Mixer with Equalizers December 2025
(Analog Electronics, Audio Signal Processing, PCB Design)

- Designed and implemented a compact three-channel analog audio mixer with independent 3-band equalizers (bass, mid, treble), utilizing low-noise op-amps, active Sallen–Key filters, and a Class-AB audio power amplifier.
- Developed multi-board PCBs and enclosure designs with noise-reduction techniques, power-supply isolation, thermal management, and hardware testing to ensure reliable low-distortion audio performance.

Automated Switch for Clothing Iron June 2025
(Embedded Systems, PCB and Enclosure Design)

- Designed and developed a smart auto-off socket for clothing irons that detects user inactivity and automatically disconnects AC power.
- Designed a PIC microcontroller–based embedded controller integrated with an opto-isolated TRIAC power stage for automated and safe switching of high-voltage AC loads.

PWM-Controlled DC-DC Buck Converter May 2026
(Power Electronics, Control Systems, Analog Electronics)

- Designed and implemented a PWM-controlled DC–DC buck converter (15 V input, 3–12 V adjustable output, 1 A load current) including MOSFET switching stage, LC output filter, PWM generation circuitry and closed-loop feedback compensator for output voltage regulation.
- Analyzed converter performance using load regulation, line regulation, and efficiency testing.

Local Area Network Design April 2026
(Computer Networks, Network Design, OSPF Routing)

- Designed and simulated a scalable university campus network and departmental LAN using Cisco Packet Tracer, implementing a three-tier architecture with VLAN segmentation, IPv4 addressing, DHCP, OSPF routing, and Spanning Tree Protocol (STP).
- Developed a fault-tolerant hybrid mesh-ring network topology for the University of Moratuwa, incorporating redundancy, inter-VLAN routing, and wireless access infrastructure with successful connectivity and routing verification through simulation.

References

1. Dr. Upeka Premaratne
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